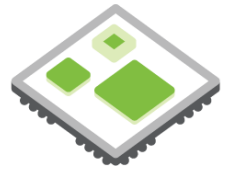
SERVER:
OPEN CHIPLET
ECONOMY



Die-to-Die Solutions for Optical Chiplets

Kevin Donnelly

Vice President, Eliyan Corporation

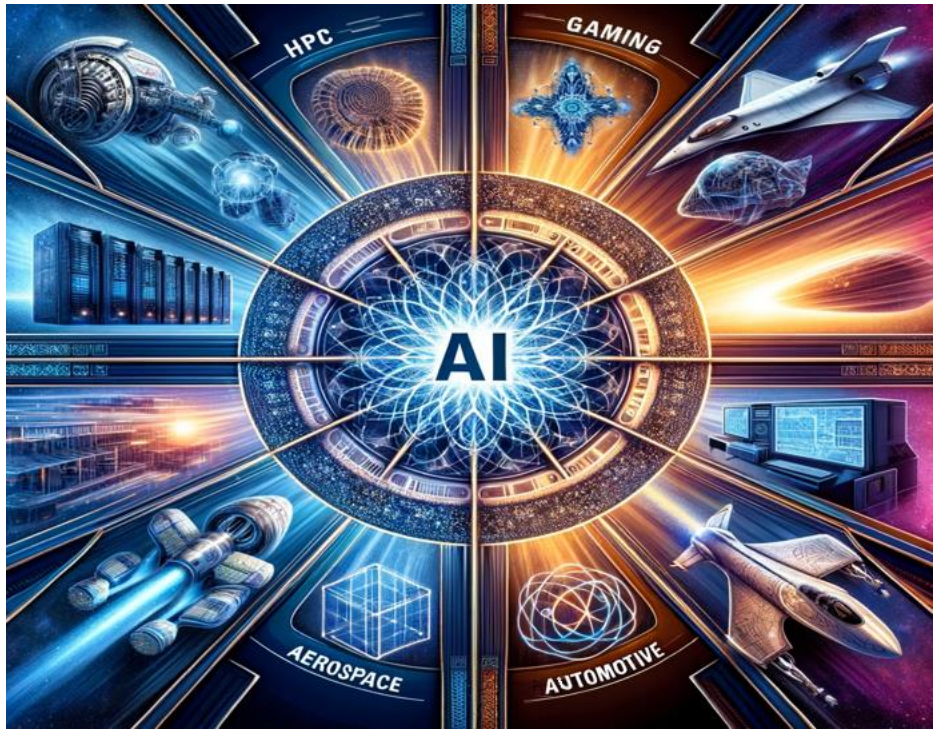
OCP BRONZE™ Member

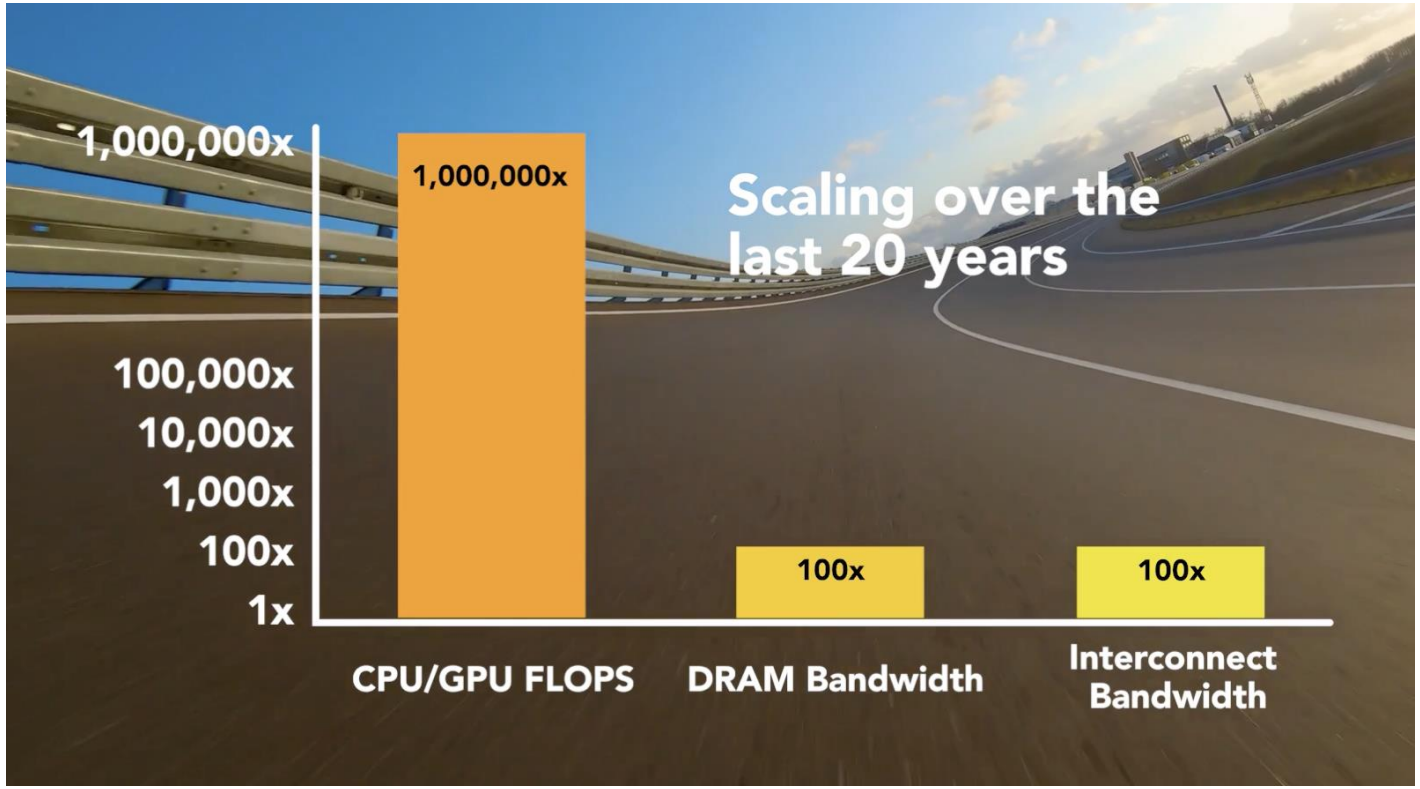
OCP INNOVATION VILLAGE

Agenda

- Background on optical modules and requirements for D2D interconnects
- Example optical modules which incorporate D2D links:
  1) Silicon Photonics with UCIe
  2) Slow-and-Wide optics with BoW
  3) Co-Packaged optics with 3D COUPE
- Summary

Top Bottlenecks in AI Performance: Memory & I/O



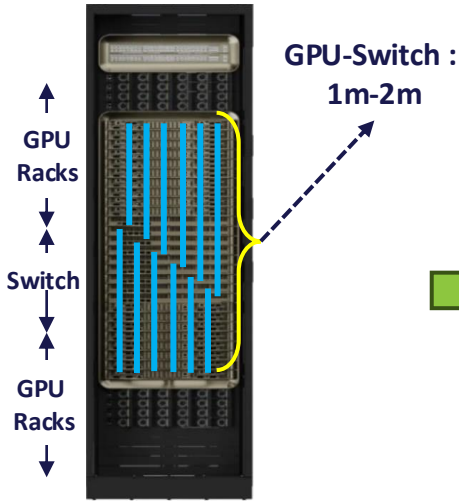


Memory & I/O Wall Bottleneck

Copper to Optical Transition

TODAY

72-288 GPU AI Pod



GPU-Switch : 1m-2m

TOMORROW

~1,000-10,000 GPU AI Pod

GPU-Switch: 10m-100m



- Next-gen AI pods (1k-10k GPUs) not only need higher bandwidth per lane (>400G/lane), but also longer connection reach (>10m)
- There is no practical copper PHY solution >400G/lane and/or >10m
- Need: A high-bandwidth optical interconnect with similar reliability, supply availability, cost and power of copper link

Reach	200Gbps	400Gbps	800Gbps+
~1m	Passive Cu	Passive Cu	Optical
~2m	Passive Cu	Active Cu	Optical
~4m	Active Cu	Optical	Optical
10m-100m	Optical	Optical	Optical

Pluggable Optical Module Solutions

- Optical modules are often placed on a common PCB near to the ASIC
- Multiple die-to-die interfaces can be used between chiplets in the modules



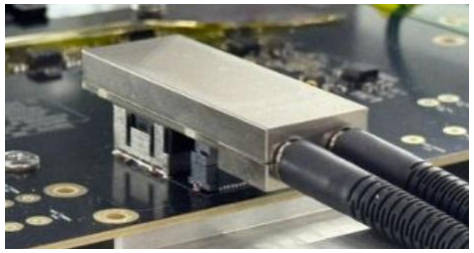
LPO



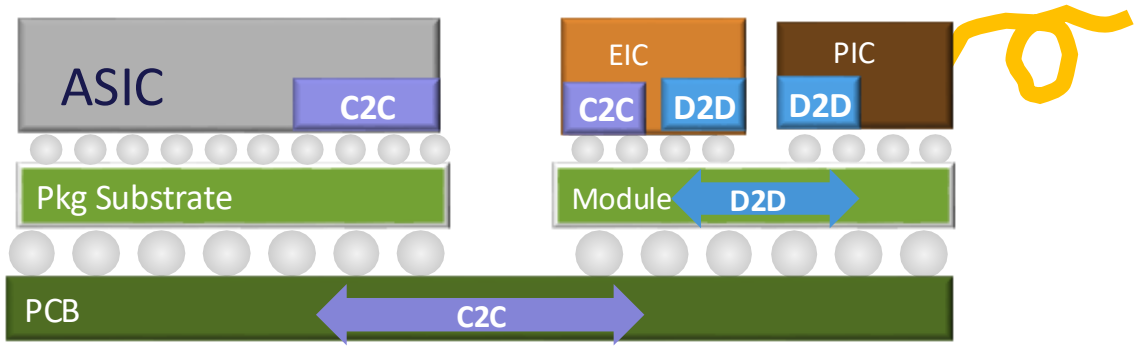
OSFP



XPO



NPO



Optical module on PCB near ASIC

Co-Packaged Optical Module Solutions

- Co-Packaged optical solutions combine optical chiplets in the same package as the ASIC connected over die-to-die interfaces



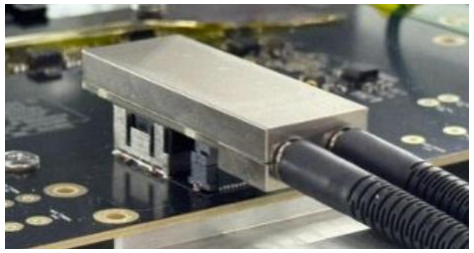
LPO



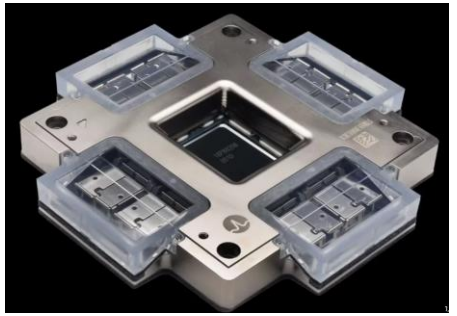
OSFP



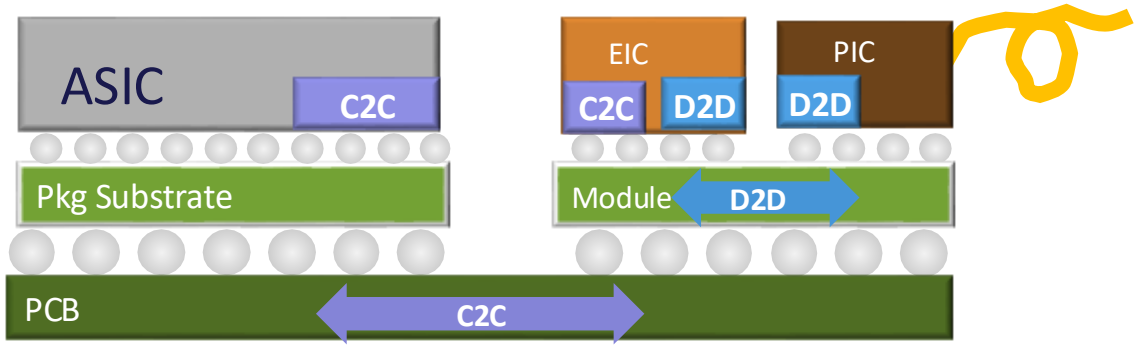
XPO



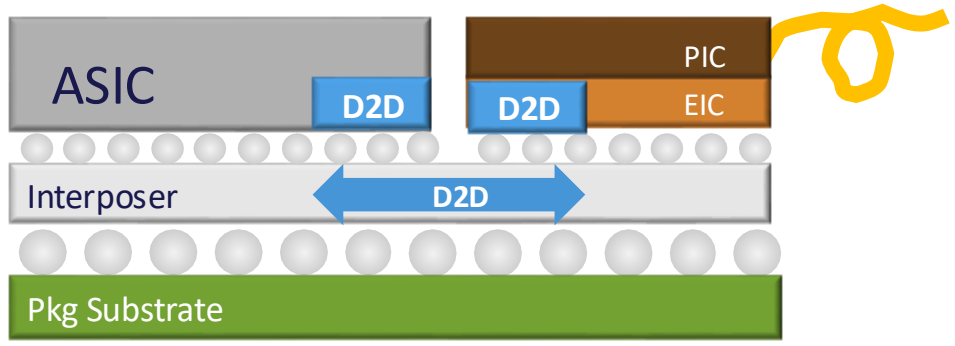
NPO



CPO



Optical module on PCB near ASIC

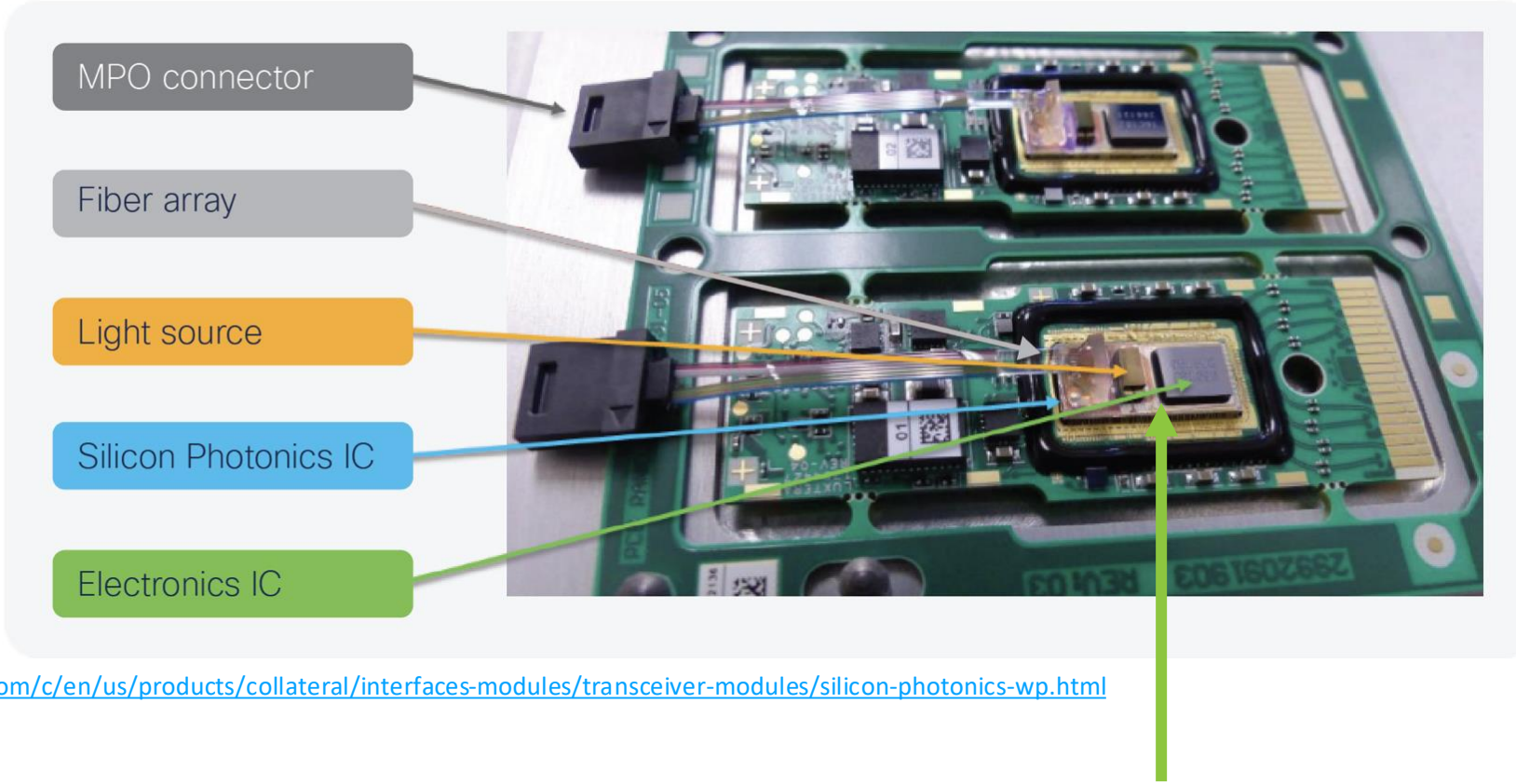


Optics co-packaged with ASIC

1) Silicon Photonics (SiPho) Optical Solutions

SiPho elements include:

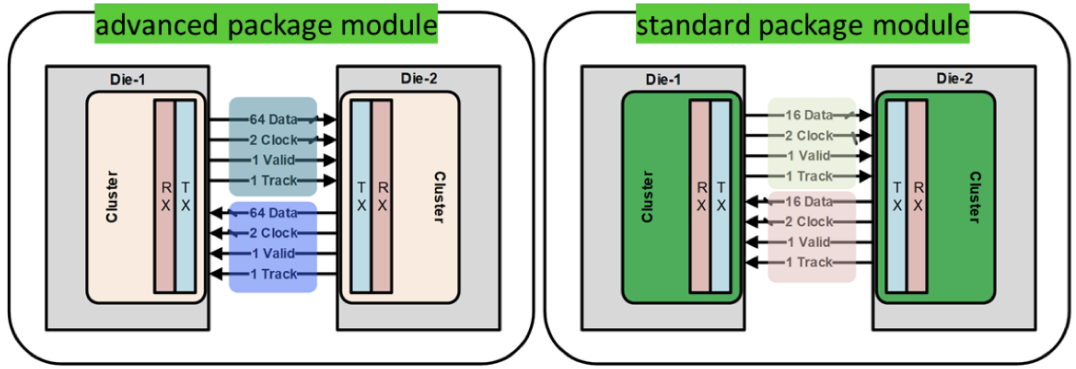
- Light source (ex Laser)
- Optical Tx/Rx (SiPho IC or PIC)
- Electrical Tx/Rx IC (EIC)
- Fibers and connectors

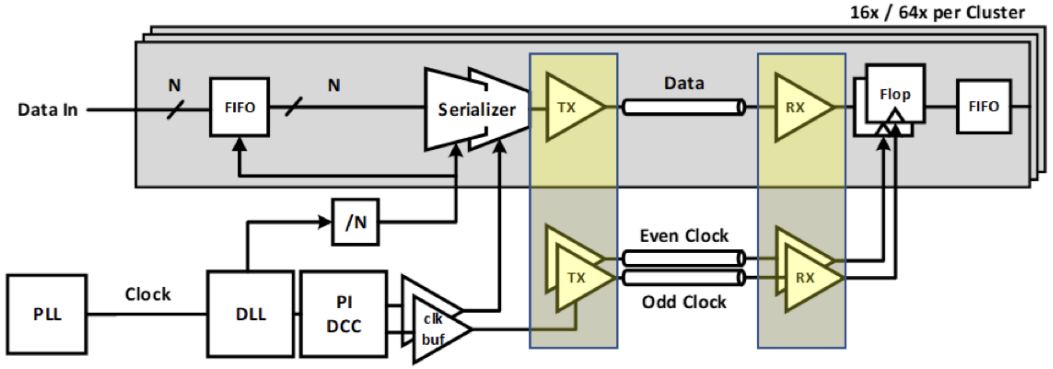


https://www.cisco.com/c/en/us/products/collateral/interfaces-modules/transceiver-modules/silicon-photonics-wp.html

D2D typically used between PIC and EIC

UCIe D2D PHY Details





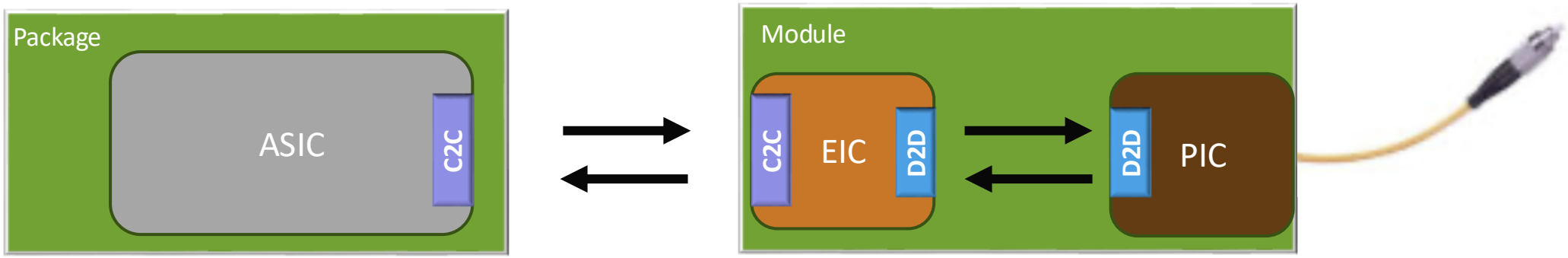
Matched Circuit Arch
Matched Interconnect

UCIe™ Universal Chiplet Interconnect Express

- Single-ended NRZ data signals
- Parallel bus with forwarded clock
  - Similar to DDR and BoW PHYs
- A standard UCIe PHY has 16-64 data lanes + forwarded clock + control signals in each direction
- Always designed with equal BW in each direction (required)
- Must be symmetric (same # RX & TX)
- Cannot have non-standard # data lanes

https://www.uciexpress.org/_files/ugd/0c1418_0c0512d5ceb342ab9499f4fbfd25feb0.pdf

UCIe D2D PHY for 1.6T SiPho Optical Solution



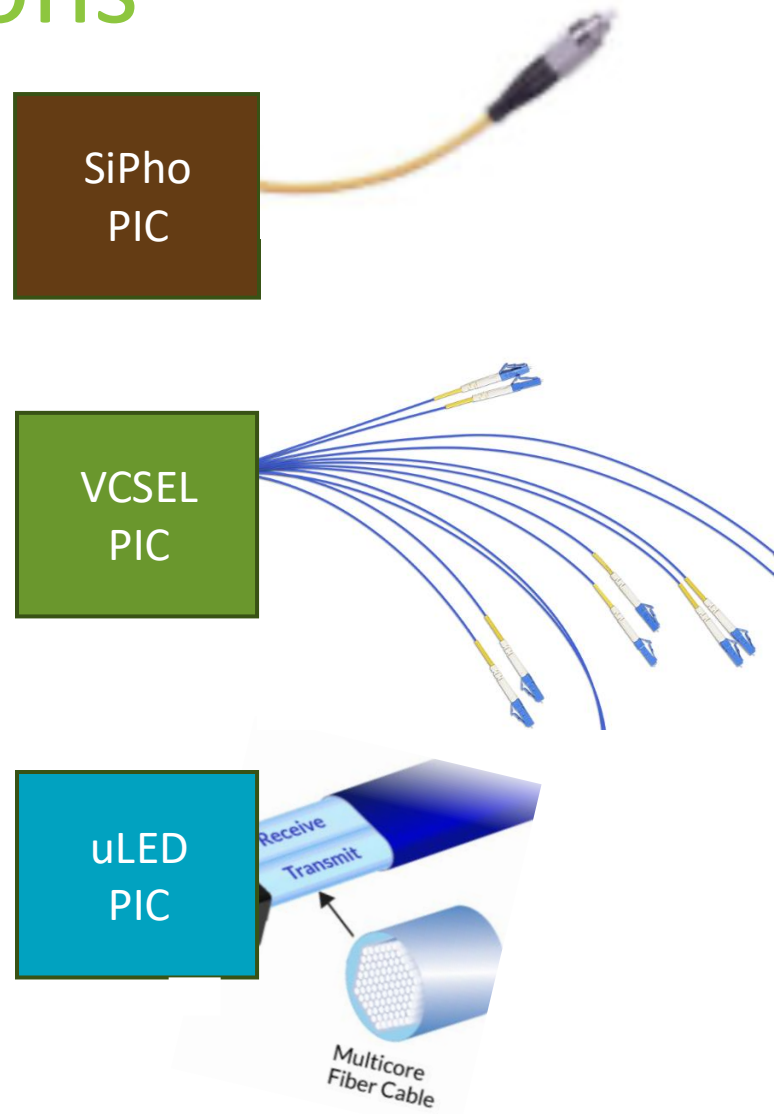
C2C
Octal 212.5G SerDes
= 1.6Tbps
(w/ Ethernet FEC overhead)

D2D
UCIe-AP x64
64 x 25.6Gbps
= 1.6Tbps
(w/ Ethernet FEC overhead)
Single clock PPM

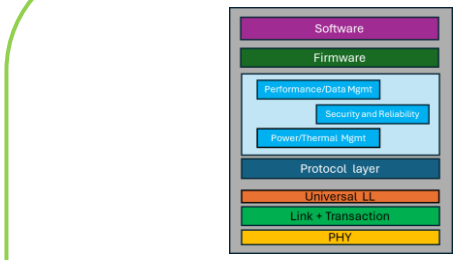
Optical
1 Fiber
1.6Tbps

2) Slow-and-Wide Optical Solutions

- Typical SiPho modules utilizing lasers drive a single fiber at very high speeds of 200G-1.6Tbps per fiber
- Slow-and-Wide solutions employ multiple optical elements such as VCSELs to drive multiple fibers as lower speeds ~25-50Gbps
- Ultra Slow-and-Wide solutions employ larger numbers of optical elements like microLEDs to drive bundles of fibers at very low speeds ~2-8Gbps

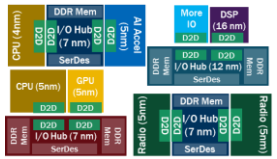


OCP Open Chiplet Economy (OCE)



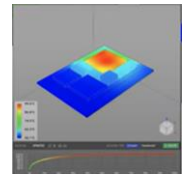
System Stacks

AI & HPC Modularity



Algorithm Decomposition

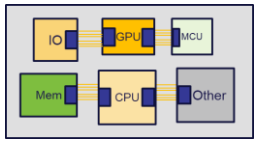




Automation for Chiplet Design

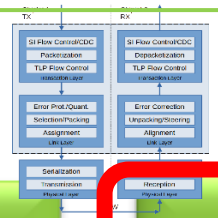
Virtual Prototyping

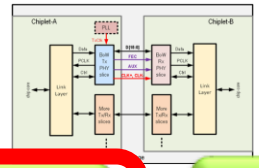
Design Tools



Foundation Chiplet Systems Architecture

Omni Connect





Logical Interconnects

BoW PHY

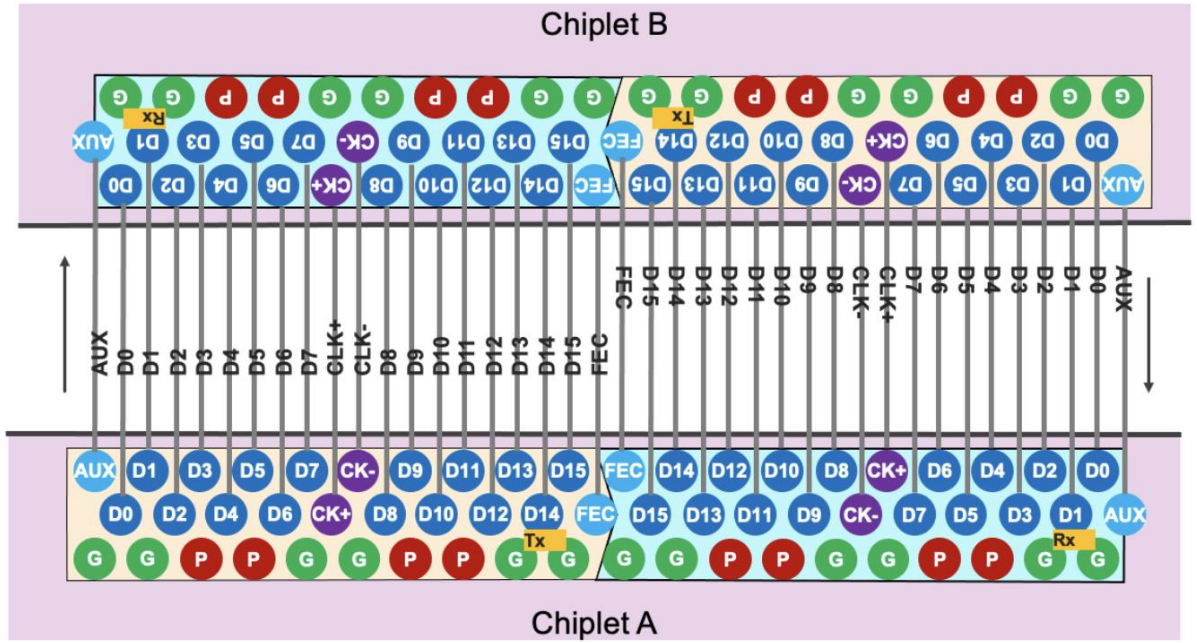
Optical Interconnects

Interconnects (D2D & Optical)

Bunch of Wires (BoW) PHY D2D Solution

- BoW provides open PHY and Link Layer specifications standardized under OCP, enabling D2D interfaces to be optimized to their host chiplet products
- The BoW ecosystem has broad industry adoption with known PHY implementations in 65nm, 22nm, 16nm, 12nm, 7nm, 6nm, 5nm, 4nm, 3nm
- Recent focus of the BoW workstream has been on spec extensions that enable additional chiplet products / product categories:
  - BoW Memory – Adding dynamic bidirectional/mixed direction slices for memory chiplets
  - BoW FlexI – SDR, streamlined PHY for cost-sensitive and/or ultra-wide applications
  - BoW Optical – Common set of recommended configurations for optical chiplets

BoW D2D PHY Details

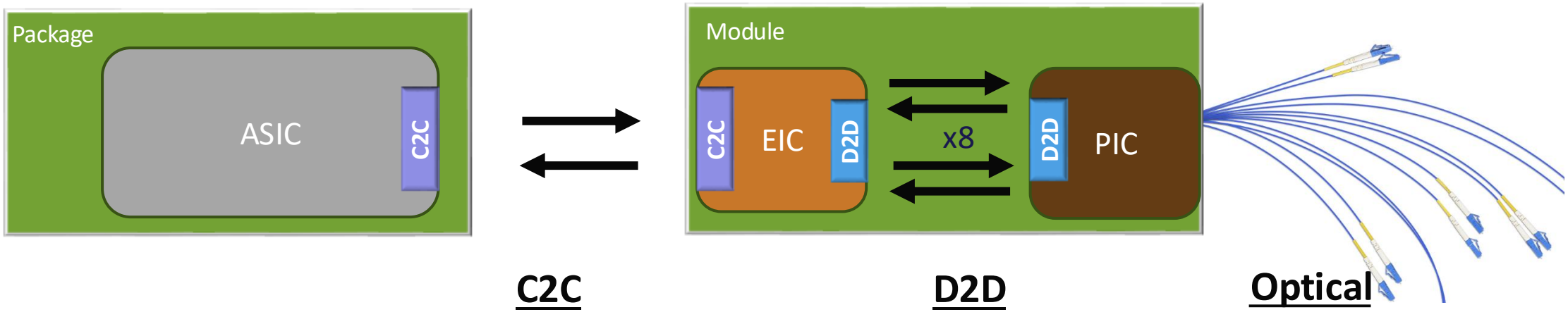


https://www.opencompute.org/documents/bow-specification-v2-0d-final-pdf

- Single-ended NRZ data signals
- Parallel bus with forwarded clock
- Similar to DDR and UCIe PHYs

- A standard BoW PHY has 16 data lanes + forwarded clock + optional AUX/FEC signals in each direction
- Typically designed with equal BW in each direction (but not required)
- Can be asymmetric (more RX or TX)
- Can be greater or fewer # data lanes
  - 32 data lanes per clock common
  - As few as 4 data lanes per clock has been implemented

BoW D2D PHY for 1.6T Slow-and-Wide Solution



C2C
Octal 212.5G SerDes
= 1.6Tbps
(w/ Ethernet FEC overhead)

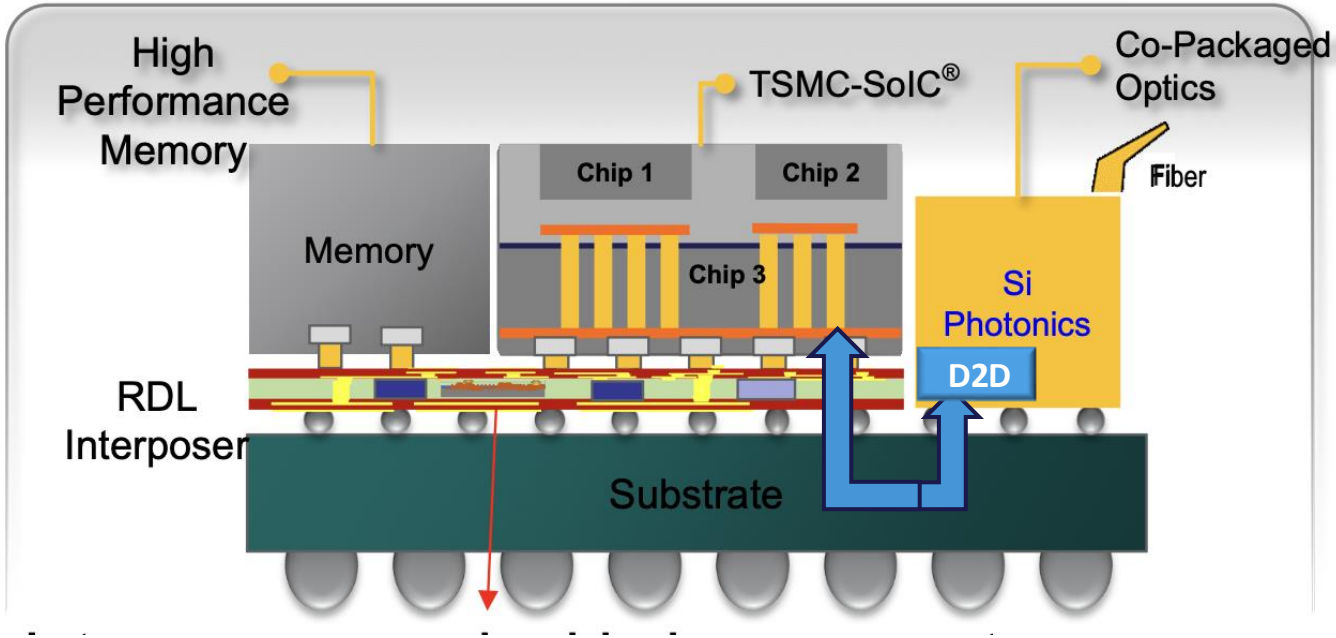
D2D
BoW-SP x4
4 x 53.1Gbps
= 200Gbps
(w/ Ethernet FEC overhead)

Eight BoW-SP x4 D2D PHYs
= 1.6Tbps with 8 separate
clock PPM

Optical
Hi Radix
8 Fiber groups at
200Gbps each
= 1.6T total

3) Co-Packaged Optics Solutions with D2D

- CPO solutions utilize D2D (not C2C)
  - Connections between chiplets within the same package substrate
- TSMC's COUPE technology is a popular choice for CPO



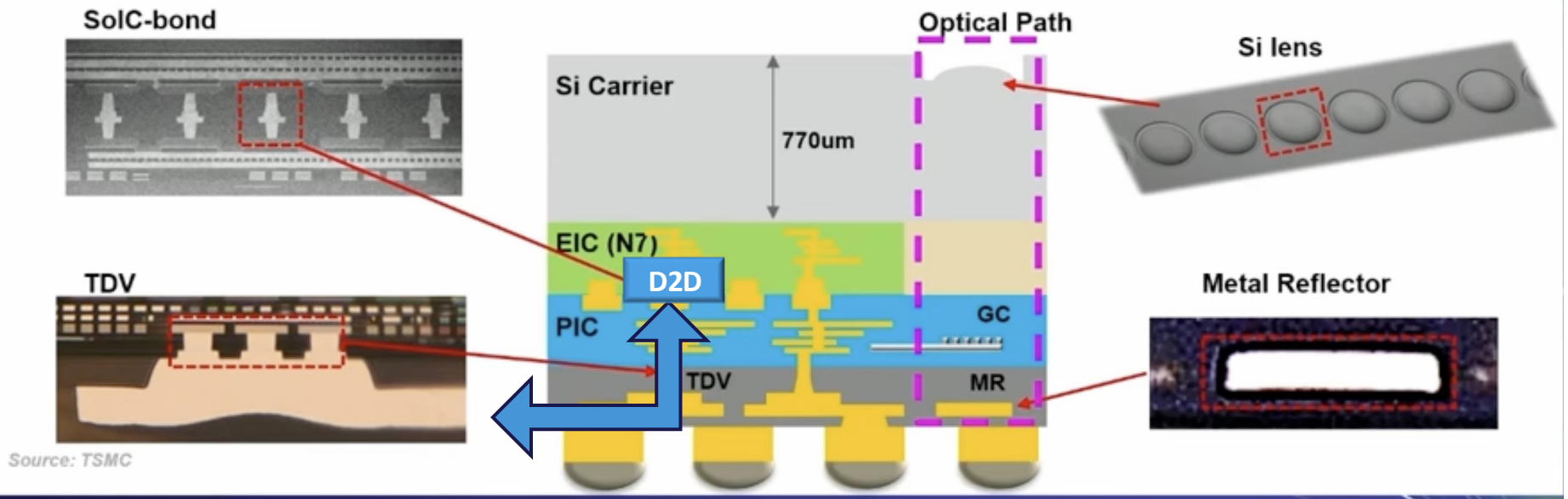
- From TSMC's keynote at OCP APAC Summit 2025: "*Advanced CPO Integrated by CoWoS and COUPE*"
- Added highlight of 2.5D D2D connection from Si Photonics through Pkg to ASIC

COUPE

- D2D connects from top EIC down through PIC (with TSVs) onto substrate
- D2D then connects in substrate (2.5D) to adjacent ASIC

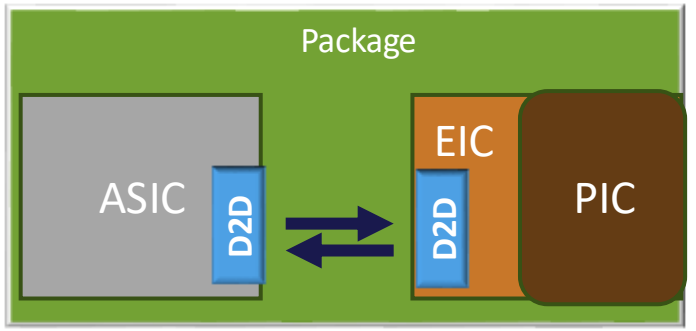
COUPE Structural Features

- Si lens is processed on a silicon carrier, and the metal reflectors are designed directly underneath the grating couplers (GC)
- ARC layers are designed in the optical path for optimum optical performance.



- From TSMC's keynote at OCP APAC Summit 2025: *"Advanced CPO Integrated by CoWoS and COUPE"*

D2D Options for CPO Solutions



- UCIe is a popular option
  - Good fit if EIC includes FEC termination and retimer functions
  - Or for EIC supporting single-fiber connections
- BoW is an excellent choice for high-radix applications
  - Best fit for multiple fiber connections to direct-drive EIC without retimer functions
  - Flexibility of lower data-to-clock ratios enables support for separate D2D clock PPMs
- 3D D2D
  - UCIe-3D and BoW-FlexI both support low-power direct-bond chiplet connections
  - Requires co-design of EIC and PIC to match bonding bump maps

Call to Action

- Problem to solve
  - New OCP-OCE optical workstream being formed - can include optimized D2D interfaces
- How to get involved in the project/sub-project community
  - OCP-OCE Wiki : https://www.opencompute.org/wiki/Server/OpenChipletEconomy
  - Wiki for new OCP-OCE optical work group : < Coming Soon >
- Additional information:
  - UCIe D2D : https://www.uciexpress.org
  - BoW D2D : https://www.opencompute.org/documents/bow-specification-v2-0d-final-pdf
  - SROI : https://www.opencompute.org/documents/sroi-white-paper-1-0-0-pdf
    [SROI = Short Reach Optical Interfaces]

Thank You!



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Connected Intelligence